

3 Introduction to Mechanics

In this chapter, we cover the mechanical aspects of making robot, including selection of the robot building material from the choices available, mounting of motors and other peripherals on the body.

3.1 Centre of mass

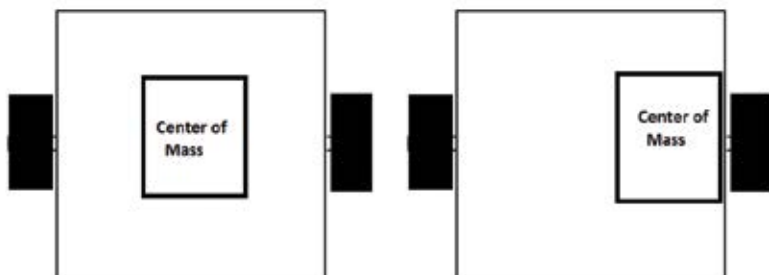
Centre of mass of a body is a point at which you can assume the whole mass of the body to be situated for analytical purposes. There are two centre of masses defined for a body: horizontal centre of mass and vertical centre of mass.

Horizontal centre of mass

As we all know, symmetry increases stability. If most of the weight of your robot is not located at the centre but concentrated to one side, the robot will be unstable. This may occur when you place your batteries, which are generally the heaviest components of your robot on one side and not at the centre. For most stable configurations, the horizontal centre of mass should coincide with the centre of the robot's body. If there is any shift in the centre of mass from the centre of body, the robot may tip over or topple.

Also, if the weight is concentrated mostly over one wheel, the grip of that wheel will be more than that of the wheel on which no significant weight is kept, because of which the robot may not be able to move straight.

A robot with a small base and high vertical center of gravity can topple easily. So, to bring necessary stability, you can design your robot in two ways: Having low height and small area of base, or, having good height, but also



Vertical centre of gravity (or mass)